MANUELA RUNGE, PhD

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EXPERIENCE

Senior Associate 09/2023 – present
MMGH Consulting GmbH, Zürich (Switzerland)

- Provide epidemiological and quantitative expertise to global health projects, supporting evidence-informed policies, strategies, and programme planning.
- Conduct epidemiological analyses, mathematical modelling, and evidence synthesis to inform infectious disease prevention and control, particularly malaria and vaccination programmes.
- Contribute to study design, methodological development, technical quality assurance, scientific reporting, and proposal development.

Independent Consultant 01/2022 – present
Manuela Runge GmbH, remote, Winterthur (Switzerland)

- Undertake scientific activities and research collaborations in infectious disease epidemiology, mathematical modelling, and quantitative analysis.
- Develop and advice on reproducible analytical workflows, modelling tools, and scientific resources.

External Research Contractor 01/2022 –12/2025
Northwestern University, remote, Chicago (United States)

- Led development of MultiMalModPy, a Python framework integrating three state-of-the-art agent-based malaria models.
- Conducted malaria modelling analyses and developed technical deliverables with research collaborators.
- Modelled the impact of perennial malaria chemoprevention and RTS,S vaccination to inform implementation in southern Nigeria.
- Supported the development of training materials for African postdoctoral researchers learning mathematical modelling with EMOD.

Postdoctoral Researcher 01/2020 – 12/2021
Northwestern University, Chicago (United States)

- Led simulations of the impact of intermittent preventive therapy in infants in collaboration with NGOs exploring the implementation of PMC in southern Nigeria.
- Led scientific projects on COVID-19 burden forecasting and malaria intervention impact modelling in Illinois.
- Trained students and co-workers in Python and the use of high-performance computing.

Doctoral Researcher 09/2016 – 12/2019
Swiss Tropical and Public Health Institute, Basel (Switzerland)

- Developed district-level malaria intervention models to support strategic planning in Tanzania; findings informed the National Malaria Supplementary Strategic Plan 2018–2020 and resulted in three peer-reviewed publications.
- Contributed to national malaria strategy development through modelling analyses and stakeholder engagement with the National Malaria Control Programme.

Scientific Assistant 03/2016 – 08/2016
Swiss Tropical and Public Health Institute, Basel (Switzerland)

- Communicated analytical needs, progress, and key findings to the National Malaria Control Programme and NETCELL project team in Tanzania.
- Delivered malaria modelling analyses and reports for key stakeholder meetings

Student Intern 03/2015 – 08/2015
Ministry of Health and Social Welfare Tanzania, National Malaria Control Programme

- Supervised data collection and data entry.
- Led data cleaning and data analysis, leading towards a scientific publication.
- Generated a new epidemiological malaria map using malaria prevalence in school children, used to inform malaria risk stratification.

Medical Data Manager 03/2014 – 07/2016
University Medical Center Hamburg-Eppendorf, parttime, remote, Hamburg (Germany)

- Supported statisticians in their statistical guidance of clinical researchers and doctoral students.
- Generated automated data description and analysis reports shared with clients.

PROFESSIONAL ACTIVITIES

2026 – present	Associate Editor, Malaria Journal
2021 – present	Committee Member, Applied Malaria Modelling Network (AMMnet)
01/2020 – 04/2024	Senior Editor – Malaria Eradication Alliance (MESA)
04/2018 – 05/2020	Correspondent – Malaria Eradication Alliance (MESA)
08/2010 – 02/2011	Nurse Assistant – Marienhospital Stuttgart,

EDUCATION

2019 **PhD Epidemiology** – magna cum laude (very good)
Swiss Tropical and Public Health Institute, University of Basel, Switzerland
Thesis: "*Mathematical modelling of malaria control interventions to support strategic planning in Tanzania*"

2016 **MSc Epidemiology** – 5.5 (very good)
Swiss Tropical and Public Health Institute, University of Basel, Switzerland
Thesis: "*The Use of Public Primary Schools for Malaria Surveillance in Tanzania*"

2014 **BSc Health Sciences** – 1.3 (very good)
University of Applied Sciences Hamburg, Hamburg, Germany
Thesis: "*The influence of the menopausal hormone therapy on the overall mortality – survival analysis of the control group of the MARIE study*"

SKILLS AND MEMBERSHIPS

- Languages: German (native), English (fluent)
- Statistical programming: Python, R, SQL, Stata
- Other software and computing: QGIS, Epi Info, LaTeX, high-performance computing
- Member of the Models of Infectious Disease Agent Study (MIDAS)
- Member of the Applied Malaria Modelling Network (AMMnet)

Selected Publications

Chacky F, **Runge M**, Mwingira F, Machafuko P, Wilson B, Aaron W, et al. Institutionalizing school-based malaria surveillance to strengthen subnational stratification and policy in Tanzania: lessons from a decade of national implementation (2014–2025). Malaria Journal (article ahead of print).

Runge, M.; Karimian, Z.; Kheirandish, M.; Borghi, G.; Wodniak, N.; Fahmy, K.; Mantel, C.; Cherian, T.; Nabil Ahmed Said, Z.; Najafi, F.; et al. COVID-19 Vaccine Effectiveness Studies against Symptomatic and Severe Outcomes during the Omicron Period in Four Countries in the Eastern Mediterranean Region. Vaccines 2024, 12, 906.

Toh KB**, Runge M**, Richardson RA, Hladish TJ, Gerardin J. 2023. Design of effective outpatient sentinel surveillance for COVID-19 decision-making: a modeling study. BMC Infect Dis 23, 287.

Runge, M., Richardson, R.A.K., Clay, P.A., Bell, A., Holden, T.M., Singam, M., Tsuboyama, N., Arevalo, P., Fornoff, J., Patrick, S., Ezike, N.O., Gerardin, J., 2022. Modeling robust COVID-19 intensive care unit occupancy thresholds for imposing mitigation to prevent exceeding capacities. PLOS Global Public Health 2, e0000308.

Runge M, Thawer SG, Molteni F, Chacky F, Mkude S, Mandike R, et al. Sub-national tailoring of malaria interventions in Mainland Tanzania: simulation of the impact of strata-specific intervention combinations using modelling. Malar J. 2022;21: 9.

Runge, M., Mapua, S., Nambunga, I., Smith, T.A., Chitnis, N., Okumu, F., Pothin, E., 2021. Evaluation of different deployment strategies for larviciding to control malaria: a simulation study. Malaria Journal 20, 324.

Ozodiegwu, I.D., Ambrose, M., Battle, K.E., Bever, C., Diallo, O., Galatas, B., **Runge, M**., Gerardin, J., 2021. Beyond national indicators: adapting the Demographic and Health Surveys' sampling strategies and questions to better inform subnational malaria intervention policy. Malaria Journal 20, 122.

Armstrong, E., **Runge, M.,** Gerardin, J., 2021. Identifying the measurements required to estimate rates of COVID-19 transmission, infection, and detection, using variational data assimilation. Infectious Disease Modelling 6, 133–147.

Runge, M., Molteni, F., Mandike, R., Snow, R.W., Lengeler, C., Mohamed, A., Pothin, E., 2020a. Applied mathematical modelling to inform national malaria policies, strategies and operations in Tanzania. Malaria Journal 19, 101.

Runge, M., Snow, R.W., Molteni, F., Thawer, S., Mohamed, A., Mandike, R., Giorgi, E., Macharia, P.M., Smith, T.A., Lengeler, C., Pothin, E., 2020b. Simulating the council-specific impact of anti-malaria interventions: A tool to support malaria strategic planning in Tanzania. PLoS ONE 15, e0228469.

Thawer, S.G., Chacky, F., **Runge, M**., Reaves, E., Mandike, R., Lazaro, S., Mkude, S., Rumisha, S.F., Kumalija, C., Lengeler, C., Mohamed, A., Pothin, E., Snow, R.W., Molteni, F., 2020. Sub-national stratification of malaria risk in mainland Tanzania: a simplified assembly of survey and routine data. Malaria Journal 19, 177.

Chacky, F., **Runge, M**., Rumisha, S.F., Machafuko, P., Chaki, P., Massaga, J.J., Mohamed, A., Pothin, E., Molteni, F., Snow, R.W., Lengeler, C., Mandike, R., 2018. Nationwide school malaria parasitaemia survey in public primary schools, the United Republic of Tanzania. Malaria Journal 17, 452.